AMAN ATAR

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SUMMARY

Machine Learning & Quantitative Systems Engineer pursuing an **M.Sc. in Financial Engineering at WorldQuant University**. Ranked **#30 Worldwide on Kaggle (Top 0.05%, 40 Medals)** and **Gold Level on WorldQuant BRAIN**, with verified experience shipping AWS EC2 LLM failover systems, 480-run SEC 10-K Hybrid RAG benchmarks, and 0.40 ms PyTorch Differential ML option pricers verified on Ubuntu 24.04 CI.

TECHNICAL SKILLS

Machine Learning & Quant: Python, PyTorch, scikit-learn, LightGBM, XGBoost, CatBoost, LinearSVC, Differential ML (Sobolev Loss), Conformal Prediction (CQR), Carr-Madan FFT, Bates SVJD, HMM Regime Detection, Soft-Sinkhorn Optimal Transport
GenAI, NLP & Retrieval: Retrieval-Augmented Generation (RAG), FAISS, BM25, Reciprocal Rank Fusion (RRF), Cross-Encoder Reranking, Sentence-Transformers, HuggingFace TRL (SFT), Google ADK (Multi-Agent), Amazon Bedrock, Gemini API, NVIDIA NIM, Groq, VADER
Cloud MLOps & Data Engineering: AWS EC2, Docker, Linux (Ubuntu 24.04), GitHub Actions CI/CD, MLflow, Flask, Streamlit, Gunicorn, pytest, PostgreSQL, Google BigQuery, Apache Spark, Kafka, dbt, Kestra, Terraform, SQL, Git

PROFESSIONAL EXPERIENCE

Innomatics Research Labs — **Nov 2025 – Mar 2026**
Data Science with GenAI Intern — *Hyderabad, India (Remote)*

- Deployed **FinWiser** on **AWS EC2 (Ubuntu, Gunicorn, systemd)** with a live circuit-breaker failover that intercepts Gemini API HTTP 429 rate-limit drops and routes traffic to **NVIDIA NIM (Kimi & Llama 3.1 70B)** with zero session loss.
- Built an enterprise semantic Q&A pipeline using **Amazon Bedrock Knowledge Bases** with automated document chunking, alongside a structured YouTube-transcript-to-PDF synthesis pipeline.
- Benchmarked 5 sentiment-classification models in **MLflow** on imbalanced Flipkart review data, optimizing hyperparameter sweeps on macro F1-score (82.12% confidence) and serving the winning model via Flask on EC2.

TheSmartBridge — **Nov 2025 – Jan 2026**
Machine Learning Engineer Intern — *Hyderabad, India (Remote)*

- Built an end-to-end 4-stage clinical hypertension risk stratification system (**Normal, Stage-1, Stage-2, Crisis**) served via a calibrated **Flask** web interface (github.com/Amanatar/hypertension-risk-assessment).
- Locked train-to-inference feature scaling invariants inside a single serialized preprocessing artifact to prevent distribution drift across clinical classes, verified via automated **pytest** HTTP 200 smoke suites in GitHub Actions CI.

Edunet Foundation (AICTE × IBM SkillsBuild) — **Sep 2025 – Oct 2025**
AI & Cloud Technology Intern — *Remote*

- Built **Student Mental Health Companion** (featured AICTE × IBM capstone), pairing the **Gemini API** with real-time **VADER** compound sentiment routing and deterministic offline crisis-helpline fallbacks (github.com/Amanatar/student-mental-health-companion).

SELECTED EMPIRICAL SYSTEMS & RESEARCH (10/10 UBUNTU 24.04 CI VERIFIED)

FinFaithful-RAG: 480-Run SEC EDGAR 10-K Hybrid RAG Benchmark · github.com/Amanatar/FinFaithful-RAG (CI #42760579773)

- Evaluated 4 retrieval tiers across **480 live runs** (120 audited questions over **AAPL, MSFT, NVDA, JPM, GOOGL** Form 10-K filings) using a decoupled generator (`qwen3.8-27b`) and independent judge (`gpt-oss-20b`).
- Engineered Markdown table-header replication, per-ticker quota routing, and **Hybrid RRF (BM25 + FAISS) + Cross-Encoder reranking**, lifting **HitRate@5 from 44.4% (FAISS) and 54.4% (BM25) to 72.2%** and doubling cross-company comparative recall (**33.3%** → **66.7%**).
- Raised grounded answer faithfulness from **2.66/5.0** (ungrounded baseline, 0.0% fact coverage) to **4.12/5.0** (**89.2%** inline citation validity) and achieved **100.0% abstention accuracy** across 30 adversarial hallucination traps.

Neural Bates DML: Sobolev Differential ML Option Pricing Surrogate · github.com/Amanatar/neural-bates-dml (CI #41193397993)

- Trained a twin-network **PyTorch** surrogate for the 10-parameter **Bates SVJD** jump-diffusion model on analytical **Carr-Madan FFT** option prices and exact Greeks ($\Delta, \mathcal{V}$, Jacobians), evaluating a **128-option batch in 0.40 ms on CPU** with zero NaN instability.

Kronos v1.0.0 & TRACE-Fin: Point-in-Time Conformal Econometrics · github.com/Amanatar/auditable-regime-aware-agentic-forecasting (CI #41193425227)

- Audited asymmetric tail sizing across **4,450 out-of-sample asset-days (890 NYSE sessions, 2023–2026)** net of 25 bps costs; Bowley quantile skewness sizing reduces maximum drawdown by **30% (-0.14% vs. -0.20%)** with deterministic replication ($< 3.33 \times 10^{-16}$) in **0.47s**.
- Enforced strict point-in-time cutoff invariance (`cutoff_validator.py`, 0 lookahead leakage), HMM regime detection, and calibrated **90% ($\alpha = 0.10$) conformal intervals** in **0.04s**.

Competitive ML Systems: LLM Behavioral Forensics & Gold Kernels · kaggle.com/amanatar

- Reached **0.684 Public LB (Rank 353 / 5,728, Top 6%)** in *Name That LLM* via `Q3-8B`/`WeMM-4B` embeddings, Soft-Sinkhorn transport, ATLAS subspace scoring, and 4-view `LinearSVC` (539k sparse features); built `ionwraith-v18` over 324,884 MS/MS spectra in *CASMI 2026*; authored 4 Gold notebooks led by `ans-verifys` (**749+ upvotes**).

EDUCATION

WorldQuant University — *M.Sc. in Financial Engineering (MScFE)* — **2025 – Present**
Coursework: Financial Econometrics (MScFE 610), Derivative Pricing (MScFE 620), Stochastic Modeling

Indian Institute of Technology (IIT) Hyderabad — *Coursework in AI, Machine Learning & Autonomous Systems* — **2025 – Present**

Punyashlok Ahilyadevi Holkar Solapur University — *Bachelor of Computer Applications (BCA)* — **Graduated June 2025**

HONORS & CERTIFICATIONS

Kaggle Notebooks Expert — Global Rank #30 / 58,650+ (Top 0.05%, Peak #28): 40 Medals (4 Gold, 5 Silver, 31 Bronze) & 2,800+ upvotes.
WorldQuant BRAIN Challenge — Gold Level (2025 – Present): Market-neutral, low-turnover cross-sectional equity alpha research.
DataTalks.Club Data Engineering Zoomcamp (2026 Cohort): Docker, PostgreSQL, Terraform, BigQuery, Kestra, dbt, Spark, Kafka.